

EXECUTIVE BUSINESS REPORT: DEMAND INTELLIGENCE & SALES FORECASTING

TO: Head of Supply Chain & Chief Financial Officer (CFO)
FROM: Lead Data Scientist / Demand Intelligence Intern
DATE: July 10, 2026
SUBJECT: End-to-End Sales Forecasting & Stocking Strategy Optimization

Executive Summary

This report presents a newly developed Demand Intelligence System designed to predict future product demand, detect operational anomalies, and segment our product catalog. By transitioning from reactive stocking to proactive forecasting, we aim to minimize capital tied up in excess warehouse inventory while preventing lost sales due to out-of-stock situations. Our analysis indicates that implementing this system can optimize stocking levels, improve regional fulfillment times, and flag high-risk transaction anomalies in real-time.

1. Key Findings from Exploratory Data Analysis (EDA)

Our historical sales data covers four years of daily transactions across three product categories (Technology, Furniture, and Office Supplies) and four geographic regions. Key insights include:

- **Revenue Drivers:** The Technology category generates the highest overall revenue (approx. \$836,154), followed by Furniture (\$742,000) and Office Supplies (\$719,000). Although Office Supplies represents the largest transaction volume, Technology drives our top-line growth due to high-value transactions.
- **Regional Growth Trajectory:** The West Region is our most consistent growth engine, showing stable year-over-year revenue expansion over the 4-year period. In contrast, the Central region exhibits high volatility with unpredictable swings.
- **Fulfillment Latency:** The average time between Order Date and Ship Date is consistently 3.9 days across all regions. There is no significant regional variation, indicating that logistics bottlenecks are likely systemic rather than regional.
- **Strong Seasonality:** We observe a highly consistent annual sales pattern. Across all four years, sales experience a massive spike in November and December (the holiday peak), as well as a secondary spike in September (corresponding to corporate Q3 budget cycles and back-to-school stocking). January and February are consistently the lowest sales months of the year.

2. Three-Month Sales Forecast

We compared three forecasting models (SARIMA, Facebook Prophet, and XGBoost) on a 3-month test horizon. The Facebook Prophet model performed the best with a Mean Absolute Percentage Error (MAPE) of 11.2%, making it our recommended production model.

The forecast for the upcoming quarter (Months 1-3) indicates a total expected revenue of \$242,500 with the following monthly breakdown:

- Month 1 Expected Sales: **\$72,400** (Normal range: \$61,800 to \$83,000)
- Month 2 Expected Sales: **\$81,200** (Normal range: \$68,900 to \$93,500)
- Month 3 Expected Sales: **\$88,900** (Normal range: \$74,200 to \$103,600)

Note: The range represents a 95% confidence interval. We are 95% confident that actual sales will fall within these boundaries. Supply chain planning should use the upper bound for safety stock calculations and the expected sales value for baseline financial projections.

3. Anomaly Report: Top 3 Sales Deviations & Likely Causes

Using a combination of Isolation Forest and Z-Score models, we identified weeks with highly unusual sales volumes. The top three high-confidence anomalies are:

- **Late November Spike (Week of Nov 22):** Weekly sales spiked to \$32,450 (normally \$11,500). This corresponds directly to Black Friday promotional activities.
- **Mid-December Spike (Week of Dec 15):** Weekly sales reached \$28,900. This aligns with corporate year-end procurement orders and holiday shipping deadlines.
- **Early September Spike (Week of Sep 5):** A sudden surge of \$24,100 occurred. This corresponds to high-volume back-to-school school district contract orders and corporate Q3 budget flushes.

4. Product Demand Segmentation & Stocking Strategy

We grouped our 17 product sub-categories into four distinct demand clusters using K-Means clustering based on total sales, year-over-year growth, sales volatility, and average order value.

Cluster Label	Sub-Categories	Characteristics	Recommended Stocking Strategy
High Volume, Stable Demand	Phones, Chairs, Storage, Binders	High revenue, high order count, predictable volatility.	EOQ Model: Maintain a standard 30-day safety stock. Use automated replenishment triggers.
Growing Demand	Copiers, Accessories, Paper	High YoY growth rate, increasing average transaction size.	Aggressive Buffer: Increase safety stock buffer by 25%. Monitor supply lead times weekly.

Low Volume, High Volatility	Machines, Bookcases, Supplies	Low order frequency but high average price; very unpredictable.	Just-In-Time (JIT): Hold minimal physical inventory locally. Consider drop-shipping.
Declining / Mature Demand	Envelopes, Labels, Art, Fasteners	Slow or negative growth, very low unit economics.	Inventory Reduction: Reduce stocking levels. Run seasonal clearance promotions.

5. Strategic Business Recommendations

- **Implement Dynamic Safety Stocking for Technology:** Since Technology drives our highest revenue and exhibits high volatility, we should transition from a flat safety stock model to a dynamic model that scales up 30 days prior to the September and November peaks. This will unlock up to \$45,000 in working capital during low-demand months (Jan-Feb).
- **Optimize East/West Regional Warehousing:** The West and East regions account for 65% of total sales. We recommend consolidating regional distribution centers in these areas to support faster fulfillment. Since the average fulfillment time is 3.9 days, we can establish a 'Fast-Track' fulfillment lane for High Volume, Stable Demand items to bring fulfillment under 2 days for key accounts.
- **Establish Promotional Pre-orders for Low-Volume/High-Volatility Items:** Sub-categories like 'Machines' tie up substantial capital. We should implement a pre-order or delayed-fulfillment model for corporate accounts purchasing machines, reducing the necessity of holding high-cost physical stock in our warehouses.

6. System Limitations & Risks

Black Swan Event Vulnerability: This demand intelligence system relies heavily on historical patterns. It cannot predict 'black swan' events such as sudden supplier factory shutdowns, regional port strikes, or macroeconomic shocks (e.g., pandemic disruptions or sudden tariffs). The system should be used as a decision-support tool, not a fully automated ordering agent.